C26 Carbonyl Compounds – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Aldehydes and Ketones **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Reducing agent for aldehydes and ketones				$NaBH_4$
2	Product of reducing an aldehyde				Primary alcohol
3	Product of reducing a ketone				Secondary alcohol
4	Symbol for a reducing agent in equations				[H]
5	Reagents to form a hydroxynitrile				KCN, then dilute acid
6	Product of ethanal with KCN/H^+				2-hydroxypropanenitrile
7	Why KCN is used rather than HCN				HCN is a toxic gas
8	Mechanism type for CN^- or $NaBH_4$ with C=O				Nucleophilic addition
9	Product of KCN with an aldehyde (optical activity)				Racemic mixture
10	Hazard of KCN				Toxic

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C26 Carbonyl Compounds – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 2: Carboxylic Acids and Esters **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Products of a carboxylic acid with a carbonate				Salt + CO_2 + H_2O
2	Reagents to make an ester				Carboxylic acid + alcohol, conc H_2SO_4 catalyst
3	Ester suffix				-oate
4	Ester from ethanol and ethanoic acid				Ethyl ethanoate
5	Uses of esters				Perfumes, flavourings, solvents, plasticisers
6	Products of acid hydrolysis of an ester				Carboxylic acid + alcohol
7	Products of alkaline hydrolysis of an ester				Carboxylate salt + alcohol
8	Fats and oils				Esters of glycerol (propane-1,2,3-triol)
9	Soap				Sodium or potassium salt of a long-chain fatty acid
10	Biodiesel				Methyl esters of long-chain fatty acids

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C26 Carbonyl Compounds – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 3: Acylation **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Acyl chloride functional group				–COCl
2	Acid anhydride general formula				$(RCO)_2O$
3	Acyl chloride + water				Carboxylic acid + HCl
4	Acyl chloride + alcohol				Ester + HCl
5	Acyl chloride + ammonia				Primary amide + HCl
6	Acyl chloride + primary amine				*N*-substituted amide + HCl
7	Mechanism type for acylation				Nucleophilic addition–elimination
8	Observation when an acyl chloride reacts with water				Steamy fumes (HCl)
9	Why anhydrides are used industrially				Cheaper, less corrosive, less violent, no HCl
10	Reagents to make aspirin				2-hydroxybenzoic acid + ethanoic anhydride

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.